



WeCharge

Sensorless Dynamic Wireless Charging

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The Problem

Charging is a bottleneck for anything that moves.

- Vehicles, AGVs, cranes stop moving to charge — downtime, cabling, oversized batteries.
- Existing dynamic charging (DWPT) needs position sensors + dense multi-coil arrays → costly, complex.
- Misalignment kills efficiency; strong fields raise EMF safety concerns.
- Exposed cabling is a single point of failure — a resilience gap during grid or crisis disruption.



\$16.4B

Global wireless power
transfer market, 2025

\$105B

Forecast by 2035

20.4%

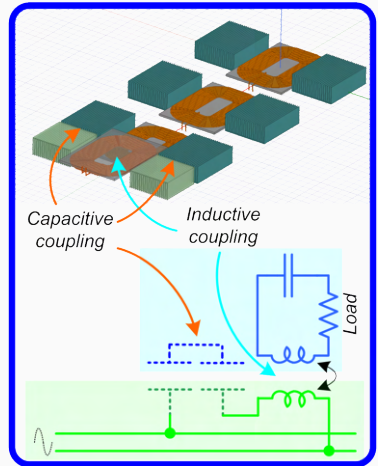
CAGR, 2025–2035

- Wireless EV charging is still an **early niche** ($\approx$ \$0.1B in 2025); dynamic (in-motion) charging exists mainly as expensive sensor-based pilots (ORNL, FastInCharge).
- Adjacent pull: industrial AGVs, cranes, public transport, defense & emergency mobility.

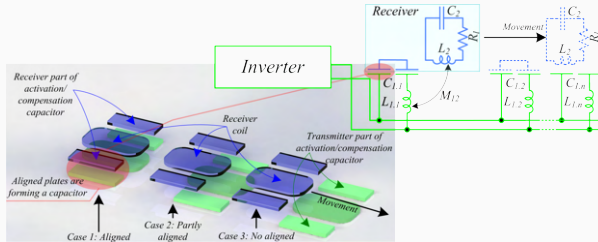
Sources: Market Research Future — Wireless Power Transmission Market (2025–2035); Precedence Research — Wireless EV Charging Market (2025–2035).

Our Solution: Hybrid Sensorless Coupling

- Combines inductive + capacitive coupling in one transmitter segment.
- The coupling capacitor doubles as a self-activating switch — it closes the current path only when the receiver overlaps the segment.
- Result: no position sensors, no photocells, no receiver-side beacon. Physics activates the right coil automatically.
- Patent-protected core mechanism.



How It Works



Sensorless — only current-phase measurement needed to detect activation.

Low EMF — capacitive coupling partially shields stray field.

Less copper — high-frequency (up to 1 MHz) GaN converters shrink coils.

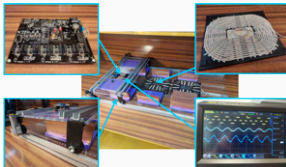
Scales — multi-phase track segments, limited only by bus losses.

	Conventional DWPT	WeCharge
Activation	Position sensors	Sensorless
Structure	Multi-coil array	Single hybrid segment
EMF exposure	High, shielded	Reduced (capacitive)
Control	High (array switching)	Low (phase only)
IP position	Open / institutional	Patent LV 15706 B1



Patent LV 15706 B1

Sensorless transmitter activation



Working Prototype

Built & tested, low-power



WPT Conference 2022

Bordeaux, France

Team track record: 11 journal papers, 22 conference papers, 5 patents, 2 books.

Non-dilutive R&D funding secured

4.48M UAH project (2026–2028): theory → simulation → prototype → validation.

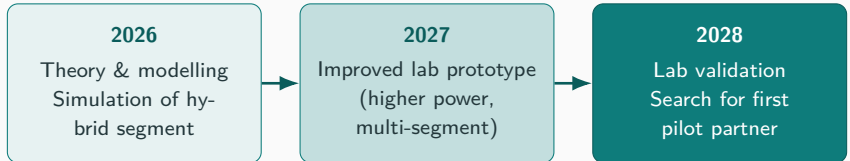
Who pays us, and for what

- **IP licensing** of the hybrid sensorless coupling design to EV charging OEMs and integrators.
- **Engineering / pilot-deployment contracts** — logistics & AGV fleets, overhead cranes, factory automation.
- **Public transport pilots** (buses, monorail) via infrastructure partners — lower EMF and no exposed cabling ease regulatory approval.

Target performance

Efficiency up to 90% · 10–20% smaller/lighter coils · up to 30% lower idle losses — validated at prototype scale, being scaled under the current funded project.

Roadmap



Team



Pakhaliuk Bohdan



Shevchenko Viktor



Yatchenko Yevhenii



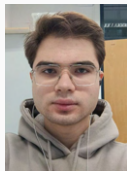
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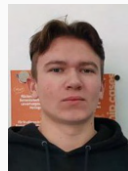
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Baida Vladyslav



Lysiuk Vladyslav



Zhuk Dmytro

What we're seeking

- **Mentorship** on commercialization and market validation.
- **Contacts with industrial companies** (logistics, cranes, transport) interested in future testing.
- **Advice** on international patent protection.
- **Small-scale support** for prototype components and lab testing.

Let's charge things without ever plugging in.

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